AI-ENHANCED SMART WOMEN’S SAFETY DEVICE USING IOT.

DEVS REC BOARD RECRUITMENT 2026-ROUND 2

Submitted by Christy Grace V.
B.E.CSE-CS, REC

1.Introduction

Women's safety remains one of the most important concerns in modern society. During emergencies, victims may not be able to use a mobile phone or contact emergency services in time. Delays in requesting help can increase the severity of the situation.

The **AI-Enhanced Smart Women's Safety Device Using IoT** is designed to provide a reliable and intelligent safety solution. The device combines IoT technology with automatic emergency detection to provide immediate assistance. It supports both manual activation using an SOS button and automatic detection of abnormal movements using sensors. Once an emergency is identified, the system shares the user's live GPS location with trusted contacts through the internet, enabling a faster response.

2.Problem Statement

Problem Being Addressed

Women often encounter unsafe situations while travelling alone, working late, or visiting unfamiliar locations. In many cases, they may not have enough time or the ability to manually request help. Existing safety systems rely heavily on manual activation, making them less effective during critical situations.

Target Users

- Women of all age groups
- College and school students
- Working professionals
- Elderly women
- Solo travellers

Motivation

The motivation behind this project is to enhance women's personal safety through an intelligent IoT-based device capable of automatically detecting emergencies and instantly notifying trusted contacts. The project aims to reduce response time and provide users with greater confidence and security.

3.Proposed Solution

Overview

The proposed system is built around an ESP32 microcontroller integrated with a GPS module, MPU6050 accelerometer, SOS button, buzzer, and Wi-Fi connectivity. The device continuously monitors the user's movements and responds to emergencies either manually or automatically.

When the user presses the SOS button or the sensor detects an abnormal fall, the ESP32 retrieves the user's current GPS location and sends an emergency alert through the cloud to predefined contacts.

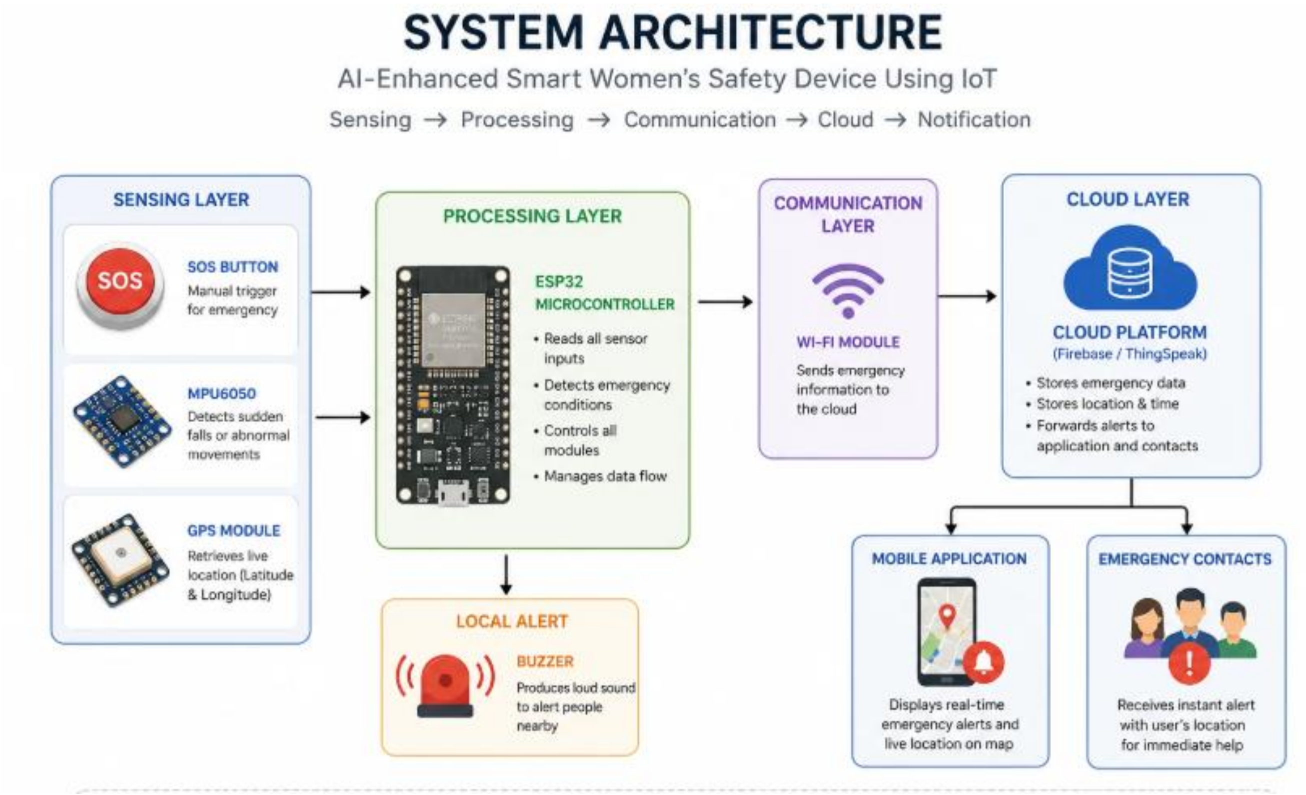
Objectives

- Improve women's personal safety.
- Detect emergencies manually and automatically.
- Share live GPS location with trusted contacts.
- Reduce emergency response time.
- Provide an affordable and portable IoT safety solution.

Expected Outcome

The proposed system will provide immediate emergency notifications, accurate location tracking, and faster assistance during emergencies. It is expected to improve user confidence and enhance overall personal safety.

4.SYSTEM ARCHITECTURE



System Architecture Explanation

The proposed system consists of an **ESP32 microcontroller**, **MPU6050 sensor**, **GPS module**, **SOS button**, **Wi-Fi**, **cloud platform**, **mobile application**, and **buzzer**. The device continuously monitors the user's safety by detecting manual SOS activation or abnormal falls. When an emergency is identified, the ESP32 retrieves the user's live GPS location and sends an alert through Wi-Fi to the cloud. The cloud then notifies the registered emergency contacts and updates the mobile application with the user's location. Simultaneously, the buzzer is activated to attract nearby attention, enabling a faster emergency response.

5.Hardware & Software Requirements

Hardware Components

- ESP32 Development Board
- Neo-6M GPS Module
- MPU6050 Accelerometer & Gyroscope
- SOS Push Button
- Buzzer
- Rechargeable Battery

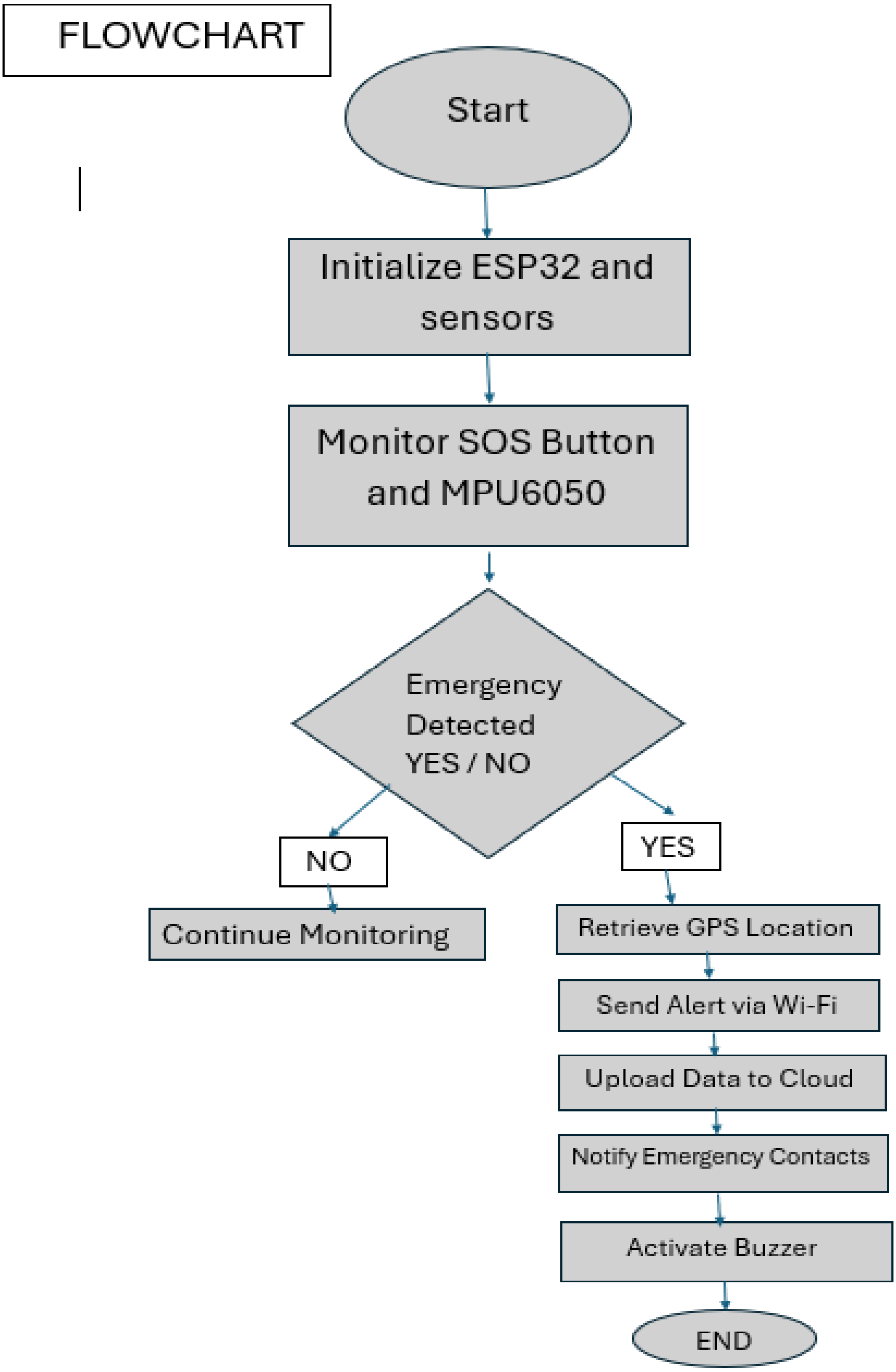
Software Components

- Arduino IDE
- Embedded C/C++
- Firebase or ThingSpeak
- Google Maps API (Optional)

6.Working Methodology

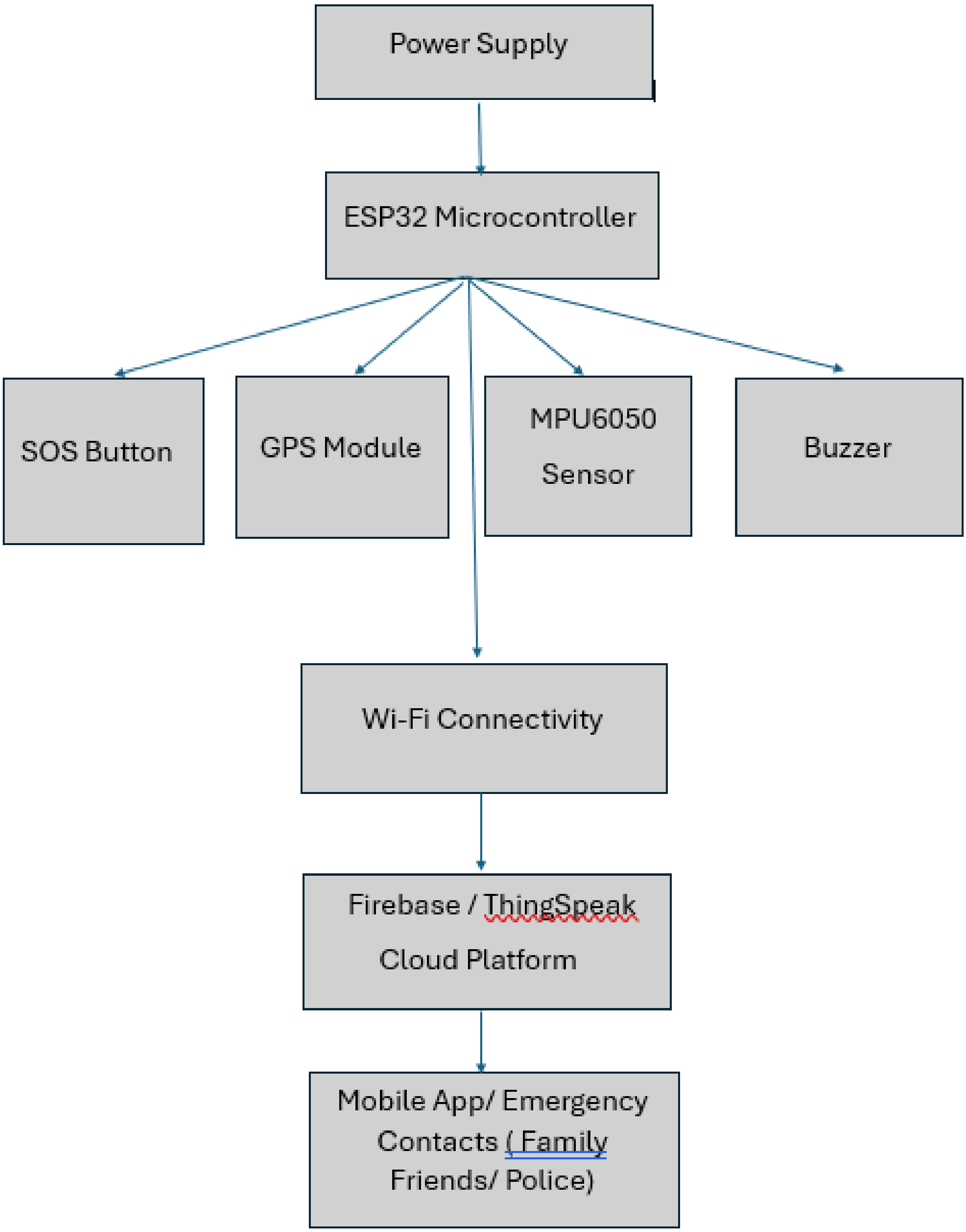
1. The ESP32 initializes all connected modules.
2. The system continuously monitors the SOS button and accelerometer.
3. If the SOS button is pressed, an emergency request is generated immediately.
4. If the MPU6050 detects a sudden fall or abnormal movement, the ESP32 verifies the event.
5. The GPS module retrieves the user's current location.
6. The emergency message and GPS coordinates are transmitted through Wi-Fi.
7. The cloud platform stores the alert and updates the dashboard.
8. Emergency contacts receive the alert with the user's location.
9. The buzzer is activated to alert nearby people.

7.

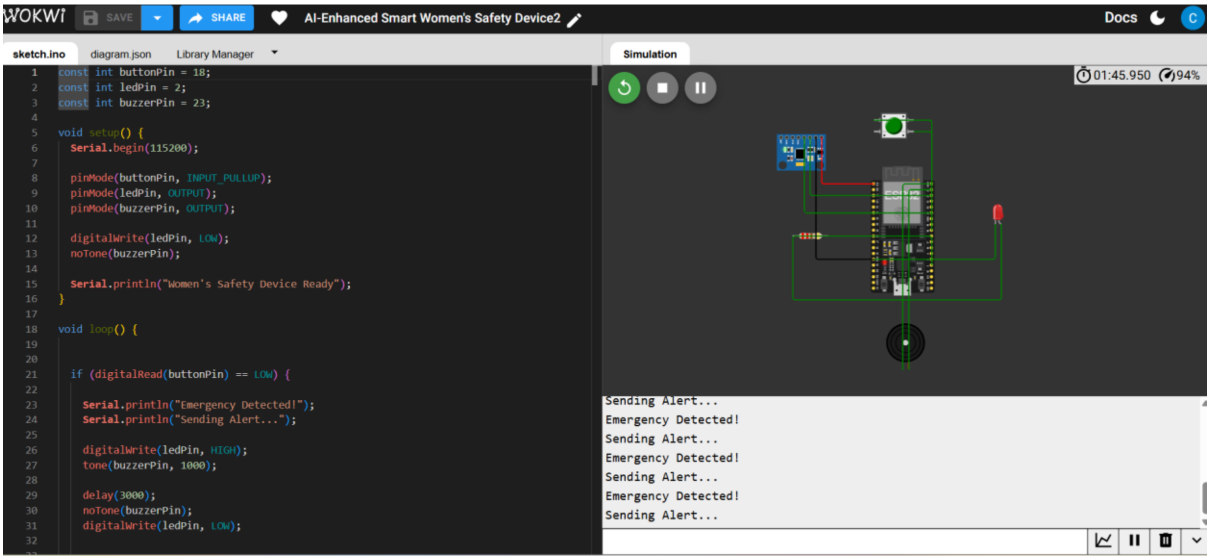


FLOWCHART
Start
Initialize ESP32 and sensors
Monitor SOS Button and MPU6050
Emergency Detected YES / NO
NO
Continue Monitoring
YES
Retrieve GPS Location
Send Alert via Wi-Fi
Upload Data to Cloud
Notify Emergency Contacts
Activate Buzzer
END

Block Diagram



9. PROTOTYPE



Prototype Simulation:

A prototype of the AI-Enhanced Smart Women's Safety Device was developed using Wokwi. The simulation demonstrates the core emergency alert mechanism. When the SOS button is pressed, the ESP32 detects the emergency, activates the buzzer and LED, and displays an emergency message in the Serial Monitor. This validates the working principle of the proposed system, while advanced features such as GPS tracking, cloud communication, and mobile notifications are represented conceptually in the system architecture.

10.Feasibility Analysis

Estimated Cost

The project uses affordable and widely available IoT components, making it cost-effective for students and practical applications.

Scalability

The system can be expanded to support multiple users, mobile applications, wearable devices, and additional sensors.

Reliability

Using ESP32 with GPS and cloud connectivity provides reliable real-time emergency communication.

Deployment Considerations

The device is compact, portable, and suitable for daily use. Stable internet connectivity is required for cloud communication.

Future Scope

- Voice-activated SOS
- AI-based danger prediction
- Smartwatch integration
- Mobile application enhancements
- Integration with emergency response services

11.Conclusion

The AI-Enhanced Smart Women's Safety Device Using IoT provides an intelligent and reliable solution for improving women's personal safety. By combining IoT technology, GPS tracking, fall detection, and cloud-based emergency communication, the system enables faster emergency response and enhances user confidence. The proposed solution is affordable, scalable, and suitable for real-world implementation.

References

1. ESP32 Documentation
2. Arduino IDE Documentation
3. Firebase Documentation
4. ThingSpeak Documentation
5. Neo-6M GPS Module Documentation
6. MPU6050 Sensor Documentation